

Mathematics Stage 3 -- Year B -- Unit 33

Multiplicative Strategies, Rates & Order of Operations

KLA: Mathematics | Stage 3 | Year 6 | Term: _____ | Duration: 2-3 weeks | Teacher: _____ | Year: _____

Syllabus outcomes

Code	Outcome descriptor
MAO-WM-01	Develops understanding and fluency through exploring, applying and communicating mathematical reasoning.
MA3-MR-01	Selects and applies appropriate strategies to solve multiplication and division problems.
MA3-MR-02	Constructs and completes number sentences, applying the order of operations to calculations.
MA3-RN-01	Applies place value to represent and regroup the natural numbers to billions.
MA4-RAT / INT	Stage 4 (group A): solves problems involving rates and ratios (RAT); applies the order of operations to integers (INT).

Syllabus content points (addressed through SOLO tracker)

All content points below are covered by students working independently through the SOLO tracker. Mini masterclasses target specific outcomes based on live tracker data.

Red -- Foundation	Yellow -- On-level	Green -- Extension
Multiplicative B & place value (recall & single procedures) Regroup whole numbers in different forms. Recognise the need to agree on the order of operations; use grouping symbols (). Multiply and divide decimals by powers of 10 (relative place value). Determine why different division questions have the same answer.	Multiplicative B (applying, rates & multi-step) Multiply whole numbers up to 4 digits by 1- and 2-digit numbers using efficient strategies. Use mental strategies to multiply benchmark decimals by single-digit numbers. Solve word problems involving rates using multiplication and division. Determine the rule relating the two rows of a table. Solve real-life problems using the order of operations and grouping symbols. Identify and use inverse operations to find unknown quantities.	Extended -- Stage 4 Ratios, Rates & Integers Represent and solve real-life rate problems, converting units and examining best buys (MA4-RAT-C-01). Simplify ratios and use the unitary method to divide a quantity in a given ratio (MA4-RAT-C-01). Apply the order of operations to expressions involving integers (MA4-INT-C-01).

Teaching model

Students work independently through the Word Labs SOLO Tracker (wordlabs.app/solo) across differentiated bands -- Grow (resources), Know (practice), Show (auto-marked assessments). The teacher monitors live dashboard data and pulls small groups for targeted mini masterclasses (15-40 min) based on which outcomes have the most incomplete Show results. Remaining session time is used for incidental one-on-one and small group teaching in response to student questions and tracker data.

Working Mathematically (MAO-WM-01)

Process	How it appears in this unit
Communicating	Students describe multiplication and division strategies, rates and order-of-operations conventions using clear mathematical language.
Understanding and fluency	Students multiply large numbers, shift decimals by powers of 10, and apply grouping symbols accurately.
Reasoning	Students reason why different divisions share an answer, determine the rule of a table, and justify the order of operations.
Problem solving	Students solve multi-step word problems involving rates and grouping symbols, and (Stage 4) ratio and integer problems.

Differentiation

Support	HPGE / Extension
Students in this group:	Students in this group:

Pre-test

Date	
Format	
Band place ments / key findings	

Post-test / unit evaluation

Date	
Key findings	
Students who need further consolidation	
Next steps	

Resources

See Appendix for Resources

Word Labs SOLO Tracker: wordlabs.app/solo | NSW DoE Stage 3 Year B Unit 33 | NESA Mathematics K-10 Syllabus (2022)

Unit 33 -- Outcome Teaching Record

Find the outcome/s you taught, write your notes, sign and date. Order follows the outcome sequence, not your teaching calendar.

Co de	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
R1	Regroup whole numbers in different forms.	RN-A Whole numbers: regroup numbers in different forms MA3-RN-01	Mini Lesson 3		
R2	Recognise the need to agree on the order of operations; use grouping symbols ().	MR-B: agree on the order in which to perform operations; use grouping symbols () to indicate operations performed first MA3-MR-01 / MA3-MR-02	Mini Lesson 2		
R3	Multiply and divide decimals by powers of 10.	MR-B: compare the relative place value of digits to multiply and divide a decimal by powers of 10 MA3-MR-01	Mini Lesson 7		
R4	Determine why different division questions have the same answer.	MR-B: determine why different division questions have the same answer MA3-MR-01	Mini Lesson 5		
Y1	Multiply whole numbers up to 4 digits by 1- and 2-digit numbers using efficient strategies.	MR-B: select and use efficient strategies to multiply whole numbers of up to 4 digits by one- and 2-digit numbers MA3-MR-01	Mini Lesson 4		
Y2	Multiply benchmark decimals by single-digit numbers mentally.	MR-B: use mental strategies to multiply benchmark decimals by single-digit numbers MA3-MR-01	Mini Lesson 8		
Y3	Solve word problems involving rates using multiplication and division.	MR-B: solve word problems involving rates using multiplication and division MA3-MR-01	Suggested: pair with G1 Mini Lesson 9		
Y4	Determine the rule relating the two rows of a table.	MR-B Patterns: determine a rule describing the relationship between the bottom and top numbers in a table MA3-MR-01	Mini Lesson 1		
Y5	Solve real-life problems using the order of operations and grouping symbols.	MR-B: investigate the order of operations using real-life contexts; solve problems involving grouping symbols MA3-MR-01 / MA3-MR-02	Mini Lesson 10		
Y6	Identify and use inverse operations to find unknown quantities.	MR-B Number sentences: identify and use inverse operations to assist with the solution of number sentences MA3-MR-02	Mini Lesson 6		
G1	Represent and solve real-life rate problems (convert units, best buys).	Stage 4 RAT-A: simplified rates; convert units for rates; solve real-life rate problems incl. best buys MA4-RAT-C-01	Mini Lesson 11		
G2	Simplify ratios and use the unitary method to divide a quantity in a given ratio.	Stage 4 RAT-A: compare quantities as ratios; simplify ratios; unitary method; divide a quantity in a given ratio MA4-RAT-C-01	Mini Lesson 12		
G3	Apply the order of operations to expressions involving integers.	Stage 4 INT-A: apply the 4 operations to integers; grouping symbols; order of operations with integers MA4-INT-C-01	Mini Lesson 13		

Syllabus outcomes and content from Mathematics K-10 Syllabus (2022) copyright NSW Education Standards Authority (NESA) for and on behalf of the Crown in right of the State of New South Wales, 2024. MR = Multiplicative Relations. RN = Represents Numbers (regrouping). RAT = Ratios and Rates (Stage 4). INT = Computation with integers (Stage 4). A/B = syllabus content groups (B is the multiplicative home content for this Year-B unit).

Unit 33 -- Mini Lesson Sequence

Multiplicative Strategies, Rates & Order of Operations | Stage 3 Year B

Supplement to the SOLO Teacher Program (pages 1-2). Print pages 1-2 only for annotation. Reference these lesson plans when planning mini masterclasses or when a reviewer requires evidence of planned lesson content.

Red band -- Foundation -- Mini Lessons 2-7

Yellow band -- On-level -- Mini Lessons 1-10

Green band -- Extension -- Mini Lessons 11-13

DoE lesson links: Lessons marked DoE Lesson X draw directly from NSW DoE Mathematics Stage 3 Unit 33. Steps have been condensed to 20-40 minutes for mini masterclass delivery. Resources referenced by their original resource numbers from the DoE unit.

Mini Lesson 1 Number patterns and the rule of a table **[HANDS-ON]**

Y4 | Yellow band | DoE Lesson 1 (adapted)

Duration: 30 min | **Outcomes:** Y4 | Syllabus: MA3-MR-01, MAO-WM-01 | MR-B: determine a rule relating the two rows of a table

Linked to: DoE Lesson 1 (adapted) -- Lesson 1 -- Number patterns and the order of operations | Key resources: Table-of-values templates, string and craft sticks (fencing-post model), counters

Physical materials needed: table-of-values templates, string and craft sticks, counters

Learning intention

Students are learning to:
determine a rule describing the relationship between the bottom number and the top number in a table

Success criteria

Students can:

- * I can look across a table to see how the top number relates to the bottom number.
- * I can write the rule (e.g. posts = metres + 1, or top = bottom x 5).
- * I can use the rule to find a missing value.

Mini masterclass structure (30 min)

Activate (4 min): Model Farmer Pat's fence: one post at the start and one post every metre. Build a table of metres vs posts.

Model (10 min): From the table, determine the rule relating posts to metres. Test it on each column and use it to predict posts for 100 m.

Guided practice (12 min): Students find the rule for several tables (including multiplicative rules) and use each rule to find missing values.

Check (4 min): Exit question: find the rule for 2->10, 3->15, 5->25 and use it to find the value for 8.

Key vocabulary

rule, relationship, table of values, pattern, multiple, term, predict

Differentiation

Support: Model the problem with string and craft sticks; draw an extra row and record the first few values visually.
Extension: Create a word problem for a partner using a given table of values (e.g. barbed-wire clips and nails).

Assessment

Teaching notes (annotate here)

Students determine and use the rule of a table — links to Y4 in the SOLO Show task.

Taught as described / adapted / replaced -- note here:

Mini Lesson 2 Order of operations and grouping symbols

R2 | Red band | DoE Lessons 1 & 6 (adapted)

Duration: 25-30 min **Outcomes: R2** Syllabus: MA3-MR-01, MA3-MR-02, MAO-WM-01 | MR-B: agree on order of operations; use grouping symbols ()

Linked to: DoE Lessons 1 & 6 (adapted) -- Lessons 1 & 6 -- The order of operations and grouping symbols | Key resources: Number-sentence cards, mini whiteboards

Learning intention

Students are learning to:
recognise the need to agree on the order of operations and use grouping symbols to show which operation comes first

Success criteria

Students can:

- * I know we must agree on an order to do operations so everyone gets the same answer.
- * I can use grouping symbols () to show which operation is done first.
- * I can evaluate a simple number sentence using grouping symbols.

Mini masterclass structure (25-30 min)

Activate (4 min): Show $3 + 4 \times 5$. Ask why two people might get 35 or 23, and why we need an agreed order.

Model (10 min): Establish: do grouping symbols () first, then $\times$ and $\div$, then $+$ and $-$. Compare $3 + 4 \times 5 = 23$ with $(3 + 4) \times 5 = 35$.

Guided practice (10 min): Students evaluate number sentences with and without grouping symbols and explain what the brackets change.

Check (3 min): Exit question: evaluate $(6 + 2) \times 3$ and $6 + 2 \times 3$.

Key vocabulary

order of operations, grouping symbols, brackets, operation, evaluate, number sentence

Differentiation

Support: Provide an order-of-operations reminder card; start with one operation plus brackets.
Extension: Place grouping symbols in an expression to reach a given target value.

Assessment

Students use grouping symbols and the order of operations — links to R2 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 3 Regrouping numbers [HANDS-ON]

R1 | Red band | DoE Lesson 2 (adapted)

Duration: 20-25 min **Outcomes:** R1 Syllabus: MA3-RN-01, MAO-WM-01 | RN-A: regroup numbers in different forms

Linked to: DoE Lesson 2 (adapted) -- Lesson 2 -- Multiplication strategies and regrouping | Key resources: Place-value chart, number expander

Physical materials needed: place-value chart, number expander

Learning intention		Success criteria
Students are learning to: regroup whole numbers in different forms		Students can: * I can regroup a number by place value (e.g. $3450 = 3 \text{ thousands} + 4 \text{ hundreds} + 5 \text{ tens}$). * I can rename a number in a non-standard form (e.g. $3450 = 345 \text{ tens}$). * I can use regrouping to make a calculation easier.
Mini masterclass structure (20-25 min)		
Activate (3 min): Show 3450 on a place-value chart. Ask how many tens are in it. Model (9 min): Regroup 3450 as 3 thousands 4 hundreds 5 tens, then as 345 tens, then as 34 hundreds 50 ones. Connect to easier multiplication. Guided practice (9 min): Students regroup a set of numbers in different forms and choose a useful regrouping for a calculation. Check (3 min): Exit question: regroup 6200 in two different ways.		
Key vocabulary		Differentiation
regroup, rename, place value, thousands, hundreds, tens, non-standard form		Support: Use a place-value chart and number expander. Extension: Regroup larger numbers (to millions) and use the regrouping to simplify a multiplication.
Assessment		Teaching notes (annotate here)
Students regroup numbers in different forms — links to R1 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 4 Multiplying large numbers

Y1 | Yellow band | DoE Lessons 2, 4 & 8 (adapted)

Duration: 30-35 min **Outcomes:** Y1 Syllabus: MA3-MR-01, MAO-WM-01 | MR-B: multiply whole numbers up to 4 digits by 1- and 2-digit numbers

Linked to: DoE Lessons 2, 4 & 8 (adapted) -- Lessons 2, 4 & 8 -- Multiplication strategies | Key resources: Grid paper for the area model, place-value chart

Learning intention

Students are learning to:
select and use efficient strategies to multiply whole numbers of up to 4 digits by 1- and 2-digit numbers

Success criteria

Students can:

- * I can use the area model and the distributive property to multiply.
- * I can multiply a 4-digit number by a 1- or 2-digit number.
- * I can choose an efficient strategy for the numbers involved.

Mini masterclass structure (30-35 min)

Activate (4 min): Multiply 1234×6 by partitioning $(1000 + 200 + 30 + 4) \times 6$. Establish the distributive property.

Model (12 min): Use the area model to multiply 1234×23 (extend the grid for the extra place value), then record with the algorithm.

Guided practice (14 min): Students multiply 3- and 4-digit numbers by 1- and 2-digit numbers using the area model and algorithm, choosing efficient strategies.

Check (4 min): Exit question: find 2456×7 and 345×24 using an efficient strategy.

Key vocabulary

area model, distributive property, partition, algorithm, multiply, efficient strategy, place value

Differentiation

Support: Use the area model for up to 3 digits first; add an extra row/column for the new place value.

Extension: Fill in a partially completed area model to find a missing multiplier; use non-standard partitioning.

Assessment

Students multiply up to 4-digit numbers efficiently — links to Y1 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 5 Why divisions share an answer

R4 | Red band | DoE Lesson 2 (adapted)

Duration: 20-25 min **Outcomes:** R4 Syllabus: MA3-MR-01, MAO-WM-01 | MR-B: determine why different division questions have the same answer

Linked to: DoE Lesson 2 (adapted) -- Lesson 2 -- Division strategies | Key resources: Counters, place-value chart

Learning intention		Success criteria
Students are learning to: determine why different division questions can have the same answer		Students can: * I can see that $600 \div 20$ and $60 \div 2$ have the same answer. * I can explain why using the relationship between the numbers. * I can use this to make a division easier.
Mini masterclass structure (20-25 min)		
Activate (3 min): Ask whether $600 \div 20$ and $60 \div 2$ give the same answer, and why. Model (9 min): Show that dividing both numbers by 10 keeps the same quotient. Use this to simplify $840 \div 40$ to $84 \div 4$. Guided practice (9 min): Students find pairs of divisions with the same answer and explain the relationship, using it to simplify divisions. Check (3 min): Exit question: write a division with the same answer as $480 \div 60$ and explain.		
Key vocabulary		Differentiation
division, quotient, divisor, dividend, equivalent, simplify, relationship		Support: Use counters to model sharing for small equivalent divisions. Extension: Generate a family of divisions with the same answer and justify the pattern.
Assessment		Teaching notes (annotate here)
Students explain why different divisions share an answer — links to R4 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 6 Inverse operations and unknowns

Y6 | Yellow band | DoE Lesson 2 (adapted)

Duration: 25-30 min **Outcomes:** Y6 Syllabus: MA3-MR-02, MAO-WM-01 | MR-B: identify and use inverse operations to find unknown quantities

Linked to: DoE Lesson 2 (adapted) -- Lesson 2 -- Inverse operations and number sentences | Key resources: Number-sentence cards, balance model

Learning intention

Students are learning to:
identify and use inverse operations to find an unknown quantity in a number sentence

Success criteria

Students can:

- * I know multiplication and division are inverse operations (and so are addition and subtraction).
- * I can use the inverse to find a missing number.
- * I can check my answer by substituting it back.

Mini masterclass structure (25-30 min)

Activate (4 min): Show $7 \times \underline{\quad} = 56$. Ask how dividing (the inverse) finds the missing number.

Model (10 min): Solve $\underline{\quad} \times 8 = 96$ with $\div$, and $144 \div \underline{\quad} = 12$ with $\times$. Check each by substitution.

Guided practice (10 min): Students complete number sentences with a missing number using inverse operations and check by substituting.

Check (3 min): Exit question: find the missing number in $\underline{\quad} \div 6 = 9$.

Key vocabulary

inverse operation, multiply, divide, unknown, number sentence, substitute, check

Differentiation

Support: Use a fact-family triangle to show the inverse relationship.

Extension: Solve two-operation number sentences using inverse operations in reverse order.

Assessment

Students use inverse operations to find unknowns — links to Y6 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 7 Decimals and powers of 10

R3 | Red band | DoE Lessons 3-5 (adapted)

Duration: 25-30 min **Outcomes: R3** Syllabus: MA3-MR-01, MAO-WM-01 | MR-B: multiply and divide decimals by powers of 10

Linked to: DoE Lessons 3-5 (adapted) -- Lessons 3-5 -- Multiplying and dividing decimals by powers of 10 | Key resources: Number slider, place-value chart, decimal cards

Learning intention		Success criteria
Students are learning to: multiply and divide decimals by powers of 10 by comparing the relative place value of the digits		Students can: * I know multiplying by 10 makes each digit's place value 10 times larger (shifts left). * I can multiply a decimal by 10, 100 and 1000. * I can divide a decimal by 10, 100 and 1000.
Mini masterclass structure (25-30 min)		
Activate (4 min): Show 3.5×10 on a place-value chart. Ask what happens to each digit. Model (10 min): Multiply $3.5 \times 10 = 35$ and $0.24 \times 100 = 24$; divide $47 \div 10 = 4.7$ and $6 \div 100 = 0.06$. Stress it is a place-value shift, not 'adding zeros'. Guided practice (10 min): Students multiply and divide decimals by 10, 100 and 1000 using a number slider. Check (4 min): Exit question: $2.4 \times 100 = ?$ and $58 \div 1000 = ?$		
Key vocabulary		Differentiation
power of 10, multiply, divide, place value, shift, decimal, relative place value		Support: Use the number slider to physically shift digits; start with $\times 10$ and $\div 10$. Extension: Find two different ways to make a decimal using multiplication or division by powers of 10 (e.g. $0.85 = 8.5 \div 10 = 850 \div 1000$).
Assessment		Teaching notes (annotate here)
Students multiply and divide decimals by powers of 10 — links to R3 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 8 Multiplying benchmark decimals

Y2 | Yellow band | DoE Lessons 4-5 (adapted)

Duration: 25-30 min

Outcomes: Y2

Syllabus: MA3-MR-01, MAO-WM-01 | MR-B: use mental strategies to multiply benchmark decimals by single-digit numbers

Linked to: DoE Lessons 4-5 (adapted) -- Lessons 4-5 -- Multiplying decimals mentally | Key resources: Number lines, place-value chart

Learning intention

Students are learning to:
use mental strategies to multiply benchmark decimals by single-digit numbers

Success criteria

Students can:

- * I can multiply 0.5 by a single digit (half of the number).
- * I can multiply benchmark decimals like 0.25 and 0.1 by a single digit.
- * I can estimate and reason about the product.

Mini masterclass structure (25-30 min)

Activate (4 min): Ask what 0.5×8 is, connecting to 'half of 8'.

Model (10 min): Multiply 0.5×9 (half of 9 = 4.5), 0.1×6 (one-tenth of 6 = 0.6), 0.25×8 (a quarter of 8 = 2). Reason about each product.

Guided practice (10 min): Students multiply benchmark decimals by single-digit numbers mentally and estimate products before recording.

Check (3 min): Exit question: find 0.5×7 and 0.25×12 mentally.

Key vocabulary

benchmark decimal, half, quarter, tenth, mental strategy, product, estimate

Differentiation

Support: Say the calculation aloud using fraction language ('half of...'); find a tenth before multiplying.

Extension: Use a 10-sided die to make decimal multiplications (e.g. 5.6×9), estimate, then find the product.

Assessment

Students multiply benchmark decimals mentally — links to Y2 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 9 Rate word problems [HANDS-ON]

Y3 | Yellow band | DoE Lessons 5 & 8 (adapted)

Duration: 30-35 min **Outcomes:** Y3 Syllabus: MA3-MR-01, MAO-WM-01 | MR-B: solve word problems involving rates using multiplication and division

Linked to: DoE Lessons 5 & 8 (adapted) -- Lessons 5 & 8 -- Rate word problems | Key resources: Vertical double number line, counters

Physical materials needed: vertical double number line, counters

Learning intention

Students are learning to:
solve word problems involving rates using multiplication and division

Success criteria

Students can:

- * I can find a rate (e.g. bricks per hour) from given information.
- * I can use the rate to scale up or down using multiplication or division.
- * I can solve a rate word problem and check it makes sense.

Mini masterclass structure (30-35 min)

Activate (4 min): Rob lays 132 bricks in 2 hours. Ask how many he lays in 1 hour (the rate).

Model (12 min): Use a vertical double number line to find the unit rate (66 bricks/hour), then scale to 5 hours and back-solve how long for 264 bricks.

Guided practice (14 min): Students solve rate problems (bricks/hour, cost/item, distance/time) using a double number line and multiplication/division.

Check (4 min): Exit question: a tap fills 18 L in 3 minutes — how much in 7 minutes?

Key vocabulary

rate, per, unit rate, double number line, multiply, divide, scale, proportion

Differentiation

Support: Provide a scaffolded double number line with 15-minute intervals marked; work out the one-hour amount first.

Extension: Solve multi-step rate problems and create rate word problems for classmates.

Assessment

Students solve rate word problems — links to Y3 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 10 Order of operations in real-life problems

Y5 | Yellow band | DoE Lessons 3, 6 & 7 (adapted)

Duration: 30 min | **Outcomes: Y5** | Syllabus: MA3-MR-01, MA3-MR-02, MAO-WM-01 | MR-B: order of operations in real-life contexts; solve problems involving grouping symbols

Linked to: DoE Lessons 3, 6 & 7 (adapted) -- Lessons 3, 6 & 7 -- Order of operations and grouping symbols in context | Key resources: Number-sentence cards, 9-sided dice, mini whiteboards

Learning intention		Success criteria	
Students are learning to: investigate the order of operations in real-life contexts and solve problems involving grouping symbols		Students can: * I can write a number sentence with grouping symbols for a real-life problem. * I can solve a multi-step problem using the order of operations. * I can use grouping symbols to control the order.	
Mini masterclass structure (30 min)			
Activate (4 min): 5 plants cost \$9 each and 3 cost \$12 each. Ask how to write this as one number sentence with grouping symbols. Model (10 min): Write $(5 \times 9) + (3 \times 12)$ and evaluate using the order of operations. Then a legs problem: 6 frogs and 4 spiders $\rightarrow (6 \times 4) + (4 \times 8)$. Guided practice (12 min): Students write and solve real-life problems using grouping symbols and the order of operations; place brackets to hit a target. Check (4 min): Exit question: write and evaluate a number sentence for '4 boxes of 6 pencils and 3 loose pencils'.			
Key vocabulary		Differentiation	
order of operations, grouping symbols, brackets, number sentence, multi-step, real-life context		Support: Use counters to calculate each part; help write the number sentence from the words. Extension: Use 4 chosen digits with any operations and grouping symbols to make as many target numbers as possible.	
Assessment		Teaching notes (annotate here)	
Students solve real-life order-of-operations and grouping-symbol problems — links to Y5 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 11 Rates in real life (Stage 4)

G1 | Green band | Original mini lesson

Duration: 30-35 min

Outcomes: G1

Syllabus: MA4-RAT-C-01, MAO-WM-01 | Stage 4 RAT-A: simplified rates; convert units; real-life rates incl. best buys

Learning intention

Students are learning to:
represent and solve real-life rate problems, converting between units and comparing best buys

Success criteria

Students can:

- * I can represent given information as a simplified rate.
- * I can convert between units for a rate.
- * I can compare best buys using rates.

Mini masterclass structure (30-35 min)

Activate (4 min): A car travels 240 km in 3 hours. Ask for the rate in km per hour (the simplified rate).

Model (12 min): Simplify a rate ($240 \text{ km} / 3 \text{ h} = 80 \text{ km/h}$), convert a rate's units, and compare two best buys (e.g. \$3.00 for 500 g vs \$5.00 for 1 kg) using unit price.

Guided practice (12 min): Students simplify rates, convert rate units, and decide best buys for shopping scenarios (AUD).

Check (5 min): Exit question: which is the better buy: 6 for \$4.20 or 10 for \$6.50?

Key vocabulary

rate, unit rate, simplify, convert, best buy, unit price, per

Differentiation

Support: Provide the unit-rate step pre-worked; compare using price per item.

Extension: Solve multi-step best-buy problems and rates involving unit conversions (e.g. mL to L).

Assessment

Students represent and solve real-life rate problems — links to G1 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 12 Ratios and the unitary method (Stage 4)

G2 | Green band | Original mini lesson

Duration: 30-35 min **Outcomes:** G2 Syllabus: MA4-RAT-C-01, MAO-WM-01 | Stage 4 RAT-A: compare/simplify ratios; unitary method; divide a quantity in a ratio

Learning intention

Students are learning to:
simplify ratios and use the unitary method to divide a quantity in a given ratio

Success criteria

Students can:

- * I can write and simplify a ratio comparing two quantities.
- * I can use the unitary method to find one part.
- * I can divide a quantity in a given ratio.

Mini masterclass structure (30-35 min)

Activate (4 min): Mix 2 cups of juice to 3 cups of water. Ask how to write and simplify this ratio.

Model (12 min): Simplify a ratio (e.g. $4 : 6 = 2 : 3$). Use the unitary method to share \$40 in the ratio 3 : 5 (8 parts, 1 part = \$5).

Guided practice (12 min): Students simplify ratios and divide quantities in given ratios using the unitary method.

Check (5 min): Exit question: divide 24 lollies in the ratio 1 : 3.

Key vocabulary

ratio, simplify, part, unitary method, divide in a ratio, compare, quantity

Differentiation

Support: Use counters to model the parts; find the value of one part first.
Extension: Solve ratio problems where one part is given and the whole must be found.

Assessment

Students simplify ratios and divide quantities in a ratio — links to G2 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 13 Order of operations with integers (Stage 4) G3 Green band Original mini lesson		
Duration: 30 min	Outcomes: G3	Syllabus: MA4-INT-C-01, MAO-WM-01 Stage 4 INT-A: 4 operations with integers; grouping symbols; order of operations
Learning intention		Success criteria
Students are learning to: apply the order of operations to expressions involving integers		Students can: * I can add, subtract, multiply and divide positive and negative integers. * I can apply the order of operations with grouping symbols. * I can evaluate an expression involving integers.
Mini masterclass structure (30 min)		
Activate (4 min): Show $-3 + 5$. Then $2 - (5 - 8)$. Ask how grouping symbols and negatives change the order. Model (11 min): Apply the order of operations to integer expressions: $-4 + 2 \times 3$, $(-6 + 2) \times 5$, and $12 \div (-3) + 4$. Work through grouping symbols first. Guided practice (11 min): Students evaluate integer expressions with grouping symbols using the order of operations. Check (4 min): Exit question: evaluate $(-2 + 6) \times 3$ and $-10 + 4 \times 2$.		
Key vocabulary		Differentiation
integer, positive, negative, order of operations, grouping symbols, evaluate, expression		Support: Use a number line for adding/subtracting integers; do one operation at a time. Extension: Evaluate multi-step integer expressions with nested grouping symbols.
Assessment		Teaching notes (annotate here)
Students apply the order of operations to integer expressions — links to G3 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Unit 33 -- Resource Appendix

Every link verified. Videos open in a browser; worksheets are print-ready PDFs (see the download checklist at the end).

Type	Resource	Link
R1 -- Regroup whole numbers in different forms.		
Video	Whole Number Expanded, Word and Standard Form Math with Mr. J	https://www.youtube.com/watch?v=NOx6jIVSSpU
Video	How to Write Whole Numbers in Standard Form Math with Mr. J	https://www.youtube.com/watch?v=H6ndRKigay8
Worksheet (print)	Corbettmaths — Place Value (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf
R2 -- Recognise the need to agree on the order of operations; use grouping symbols ().		
Video	Order of Operations: A Step-By-Step Guide (PEMDAS) Math with Mr. J	https://www.youtube.com/watch?v=x41BCrYh8Kc
Video	Order of Operations (No Parentheses or Exponents) Math with Mr. J	https://www.youtube.com/watch?v=BCg1nz5k0HY
Worksheet (print)	Corbettmaths — Order of Operations (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/order-of-operations-pdf.pdf
R3 -- Multiply and divide decimals by powers of 10.		
Video	Multiplying and Dividing by Powers of 10 Math with Mr. J	https://www.youtube.com/watch?v=HH-yrNS80Cg
Video	Place Value to the Thousandths Miacademy	https://www.youtube.com/watch?v=BsZJGMRWEK0
Worksheet (print)	Corbettmaths — Dividing by 10, 100, 1000 (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/dividing-by-10-100-1000-pdf.pdf
Worksheet (print)	Corbettmaths — Place Value (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf
R4 -- Determine why different division questions have the same answer.		
Video	How to do Long Division (1-Digit Divisors) Math with Mr. J	https://www.youtube.com/watch?v=up_xKZ6GeUg
Video	Long Division: A Step-By-Step Review Math with Mr. J	https://www.youtube.com/watch?v=HJYHNxS64f0
Worksheet (print)	Corbettmaths — Division (PDF)	https://corbettmaths.com/wp-content/uploads/2018/01/division-pdf.pdf
Y1 -- Multiply whole numbers up to 4 digits by 1- and 2-digit numbers using efficient strategies.		
Video	3-Digit by 2-Digit Multiplication Math with Mr. J	https://www.youtube.com/watch?v=dNSrfoYLQZU
Video	2-Digit by 2-Digit Multiplication Math with Mr. J	https://www.youtube.com/watch?v=PZjIT9CH6bM
Video	Box Method Multiplication Math with Mr. J	https://www.youtube.com/watch?v=n3q3XzzIGSY
Worksheet (print)	Corbettmaths — Multiplication (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/multiplication-pdf.pdf
Y2 -- Multiply benchmark decimals by single-digit numbers mentally.		
Video	How to Multiply a Whole Number and a Decimal Math with Mr. J	https://www.youtube.com/watch?v=FbZr5br9ZOg
Video	Multiply a Whole Number by a Decimal Math with Mr. J	https://www.youtube.com/watch?v=tsOibhsgYoQ
Video	Multiplying Decimals Math with Mr. J	https://www.youtube.com/watch?v=cikGV-gh-ig
Worksheet (print)	Corbettmaths — Multiplying Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2018/01/multiplying-decimals-textbook-pdf.pdf
Y3 -- Solve word problems involving rates using multiplication and division.		
Video	Unit Rate Word Problem (Double Number Line) Math with Ms. M	https://www.youtube.com/watch?v=Y7koW8jkoOs
Video	Solving Unit Rate Problems Planet Nutshell	https://www.youtube.com/watch?v=LbJUUEqAlq
Worksheet (print)	Corbettmaths — Multiplication (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/multiplication-pdf.pdf
Y4 -- Determine the rule relating the two rows of a table.		
Video	Input Output Tables: Find Patterns or Rules Easy Math with Mrs. Easley	https://www.youtube.com/watch?v=q0n7FMe1CQk

Type	Resource	Link
Video	Input & Output Tables: Rules & Missing Values STEAMspirations	https://www.youtube.com/watch?v=Z7UPvsKz4Kw
Video	Input Output Tables Song: Finding the Rule NUMBEROCK	https://www.youtube.com/watch?v=d1COlcRoSgo
Y5 -- Solve real-life problems using the order of operations and grouping symbols.		
Video	Order of Operations: A Step-By-Step Guide (PEMDAS) Math with Mr. J	https://www.youtube.com/watch?v=x41BCrYh8Kc
Video	Order of Operations (No Parentheses or Exponents) Math with Mr. J	https://www.youtube.com/watch?v=BCg1nz5k0HY
Worksheet (print)	Corbettmaths — Order of Operations (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/order-of-operations-pdf.pdf
Y6 -- Identify and use inverse operations to find unknown quantities.		
Video	How Do You Solve Inverse Operations? (KS2) Twinkl	https://www.youtube.com/watch?v=enw_kJRSt4A
Video	Input & Output Tables: Rules & Missing Values STEAMspirations	https://www.youtube.com/watch?v=Z7UPvsKz4Kw
Worksheet (print)	Corbettmaths — Multiplication (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/multiplication-pdf.pdf
G1 -- Represent and solve real-life rate problems (convert units, best buys).		
Video	Unit Rate Word Problem (Double Number Line) Math with Ms. M	https://www.youtube.com/watch?v=Y7koW8jkoOs
Video	Solving Unit Rate Problems Planet Nutshell	https://www.youtube.com/watch?v=LbJUEuEqAlg
Worksheet (print)	Corbettmaths — Best Buys (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/best-buys-pdf1.pdf
G2 -- Simplify ratios and use the unitary method to divide a quantity in a given ratio.		
Video	Unitary Method Corbettmaths	https://www.youtube.com/watch?v=AXkCfkVrjK8
Video	Dividing a Quantity in a Given Ratio Mrs G's Maths	https://www.youtube.com/watch?v=LYk4Xq9P0TQ
Video	Simplifying Ratios N.I. Maths Tutor	https://www.youtube.com/watch?v=uYj5mYPLEM
Worksheet (print)	Corbettmaths — Ratio (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/ratio-pdf.pdf
G3 -- Apply the order of operations to expressions involving integers.		
Video	Order of Operations with Integers (Positives and Negatives) Math with Mr. J	https://www.youtube.com/watch?v=AtPS0pWthEo
Video	An Intro to Order of Operations with Integers Math with Mr. J	https://www.youtube.com/watch?v=efQ1XPqw0s0
Worksheet (print)	Corbettmaths — Order of Operations (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/order-of-operations-pdf.pdf
Beyond SOLO -- Stage 4 extension		
Video	Unit Rate Word Problem (Double Number Line) Math with Ms. M	https://www.youtube.com/watch?v=Y7koW8jkoOs
Video	Unitary Method Corbettmaths	https://www.youtube.com/watch?v=AXkCfkVrjK8
Video	Order of Operations with Integers Math with Mr. J	https://www.youtube.com/watch?v=AtPS0pWthEo
Worksheet (print)	Corbettmaths — Best Buys (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/best-buys-pdf1.pdf
Worksheet (print)	Corbettmaths — Ratio (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/ratio-pdf.pdf
Website	Maths is Fun — Ratios (interactive)	https://www.mathsisfun.com/numbers/ratio.html
Website	Maths is Fun — Order of Operations (PEMDAS)	https://www.mathsisfun.com/operation-order-pemdas.html

Worksheet download checklist

Download and print these PDFs before the unit (or save to your class drive). Tick each as you go.

- ☐ Corbettmaths — Place Value (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf>
- ☐ Corbettmaths — Order of Operations (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/order-of-operations-pdf.pdf>
- ☐ Corbettmaths — Dividing by 10, 100, 1000 (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/dividing-by-10-100-1000-pdf.pdf>
- ☐ Corbettmaths — Division (PDF) -- <https://corbettmaths.com/wp-content/uploads/2018/01/division-pdf.pdf>

- ☐ Corbettmaths — Multiplication (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/multiplication-pdf.pdf>
- ☐ Corbettmaths — Multiplying Decimals (PDF) -- <https://corbettmaths.com/wp-content/uploads/2018/01/multiplying-decimals-textbook-pdf.pdf>
- ☐ Corbettmaths — Best Buys (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/best-buys-pdf1.pdf>
- ☐ Corbettmaths — Ratio (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/ratio-pdf.pdf>